

Linear algebra: Exercise chapter 11

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Autumn 2024

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1. Find the singular value decomposition of the following matrices.

$$A = \begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ -1 & -2 \end{bmatrix}, \quad \text{and} \quad C = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}.$$

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2. For the matrix B in the previous exercise, find a basis for each matrix four space.
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3. Consider the following matrix: $A = \begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix}$.

- (a) Both the column space and the row space are 1-dimensional. Find a basis for both spaces.
 - (b) Why is σ_2 zero (without calculating it)?
 - (c) Without solving systems $Ax = 0$ and $A^T x = 0$, find a basis for $N(A)$ and $N(A^T)$.
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4. Assume that $A \in \mathbb{S}^{n \times n}$ is an invertible matrix.

- (a) What is the singular value decomposition of A^{-1} ?
 - (b) What are the singular values of A^{-1} ?
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5. Why doesn't the singular value decomposition of $A + I$ just use $\Sigma + I$?
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6. Considering $m \times n$ matrix A with singular decomposition $A = U\Sigma V^T$. What is the singular value decomposition of following matrices?

- (a) A^T .
 - (b) AA^T .
 - (c) $A^T A$.
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